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Technical Note

Reducing Latency in Digital Twins: A Framework for Near-Real-Time Progress and Quality Reporting

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Abstract

While Digital Twins offer transformative potential, their efficacy for real-time control is constrained by the slow data acquisition and the high computational intensity required to process raw datasets like point clouds. This paper identifies these critical bottlenecks—specifically the latency between data capture and actionable insight—and proposes a refined theoretical framework for near-real-time automated progress monitoring and quality reporting. Building on the findings of the NORMENG project and informing the subsequent AutoGreenTraC project, this research synthesizes state-of-the-art advancements in reality capture, including LIDAR, SfM-MVS, and 360-degree vision. The study highlights a fundamental divergence in stakeholder requirements: the need for millimeter-level precision in quality control versus the demand for high-velocity documentation for progress monitoring. A key innovation presented is the shift toward neural rendering techniques to bypass the computational delays of traditional photogrammetry and enable immediate on-site visualization. By structuring a tiered processing hierarchy that combines lightweight edge analysis for immediate safety and progress monitoring with asynchronous high-fidelity Digital Twin updates, the framework aims to establish a single source of truth.

Keywords: Digital Twin; automated progress monitoring; reality capture; computer vision; AI; robotics; UAS; point cloud; edge AI

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1. Introduction—The Shift from Manual to Digital and Automated

The construction industry, a cornerstone of global infrastructure development, frequently encounters challenges in project management, often stemming from inefficient progress monitoring combined with lacking and reactive quality reporting measures and rather descriptive productivity assessment techniques [1]. This inefficiency contributes to significant cost overruns, schedule delays, and a staggering $570 billion in direct rework costs annually within the trillion-dollar construction industry, alongside an additional $440 billion in indirect costs [2,3]. While the theoretical foundations of Digital Twins (DT) and Building Information Modeling (BIM)—pioneered decades ago [4,5]—promise transformative potential, their practical efficacy for near-real-time construction control has been severely constrained by two primary bottlenecks: the slow velocity of physical data acquisition, reliant on labor-intensive manual or scanning methods prone to delays and errors [1,6,7], and the high computational intensity required for processing vast, unstructured datasets like point clouds [2,8]. These substantial financial implications highlight

a critical need for advanced technological integration to enhance project oversight and operational efficiency.

This shift from manual to automated oversight is driven by the need for objective performance benchmarks, as seen in the NORMENG project. While the project utilized high-tech modalities like LIDAR and UAS photogrammetry to move away from the subjective descriptive productivity assessments, it encountered a critical reality: simply capturing data is insufficient for real-time control. The "norming" process—calculating constructed volumes and labor efficiency—was hindered by the sheer latency of processing these vast datasets, effectively turning a high-tech solution back into a slow, reactive one. Consequently, the rationale for this paper's focus on processing speed stems directly from these field experiences, where the goal of "instantaneous" norming was unmet due to current computational constraints.

It is expected that real-time monitoring technologies will deliver comprehensive operational visibility of the construction processes through an integrated digital infrastructure. Progress monitoring integrated with DT, IoT sensors, visual-based data collection processes, and technologies is expected to enable data-driven decision-making that directly impacts schedule and cost control [9]. The development of advanced, automated systems for real-time progress monitoring and quality reporting remains a significant challenge within the construction industry [6]. Traditional methods, largely reliant on manual inspections and reporting, are time-consuming and susceptible to human error, but with the lack of educated and reliable expert knowledge, the construction industry is in dire need of a switch to digital technologies [6,10].

Within the scope of this research, "near-real-time" is operationally defined as a spectrum of processing latency dictated by a hybrid edge-cloud architecture, ranging from milliseconds for critical on-site safety interventions to minutes and hours for daily progress verification. This stands in contrast to the traditional asynchronous processing and manual reporting, where the identification of issues depends on the supervisor's capability to detect issues and assess the level of completion without complex assessments or, on the other hand, analysis of vast unstructured datasets with currently available data analysis processes, which frequently require hours to days. Consequently, "reporting" in this framework encompasses a multi-modal delivery of insights tailored to these timeframes, including instantaneous automated safety alarms, rapid on-site 3D visual deviations, and continuous cloud-based DT dashboards.

Although digital processes are anticipated to provide seamless access to project data and facilitate expedited decision-making, their practical implementation frequently encounters substantial obstacles in achieving real-time data integration and generating actionable insights. This consequently impedes timely and automated progress monitoring, effective discrepancy and quality reporting, and accurate productivity assessments. Essentially, the current state of digitalization often falls short of its anticipated benefits, particularly in delivering the promised swift and precise progress monitoring capable of mitigating project schedule delays and cost overruns [7].

Currently, in the construction industry, there is a paradigm shift occurring. While construction monitoring and control are still relying on labor-intensive manual methods, they are slowly transitioning to automated, vision-based sensing technologies to improve accuracy and timeliness in project management [1], aiming to provide a single source of truth. Here, in the paper, the single source of truth is used as a presentable, reviewable, and documented result of the construction process, which can be an information provider immediately during the construction process but also further on down the line, where the construction process is not visible anymore. Still, digitalization technologies are lagging behind the expectations of their users.

This technical note functions as a bridge between empirical observation and framework development; it utilizes specific data and performance metrics from the NORMENG project—specifically involving terrestrial LIDAR, 360-degree cameras, and UASs—to advocate for a fundamental shift in how these tools are integrated through the tiered processing architecture proposed for the AutoGreenTraC project. Here, the paper synthesizes "future work" suggestions from the recent literature calling for automated, vision-based solutions and proposes a refined theoretical framework for near-real-time automated progress monitoring and quality reporting. Emphasizing rapid data acquisition and analysis to support proactive decision-making, this foundational approach will inform research efforts in the AutoGreenTraC project.

With this research paper, we aim to gather currently available relevant scientific achievements in the field to guide future research that will bring construction practices proactive decision-making capabilities and optimization of project workflows. This research explores the integration of Structure-from-Motion Multi-View Stereo (SfM-MVS), 360-degree vision, artificial intelligence, and other innovative data collection and analysis processes for near-real-time or real-time construction progress monitoring and quality reporting through digitalization of construction processes, thereby mitigating pervasive inefficiencies. The integration of these advanced technologies facilitates a more accurate, timely, and comprehensive understanding of construction project status, moving beyond traditional manual and regular digitalization methods that are prone to assumptions, errors, and significant delays in reporting [11,12].

2. Context and Motivation

2.1. NORMENG Project as Problem Identification Phase

The NORMENG project was launched to tackle this urgent requirement by developing an automated system for resource standardization in energy-efficient construction. In a situation where a standardization framework is missing, discrepancies in productivity and material consumption are to be expected. To resolve these challenges, the project comprises two primary phases: the initial one conducted in-depth industry studies to evaluate prevailing practices and devise a reliable method for estimating material quantities, workforce demands, and machinery capacities, thereby establishing the groundwork for precise, dependable, and versatile standards. Specifically, the NORMENG project aimed to transition from manual, subjective productivity assessments to objective, data-driven "norming"—the calculation of precise labor and material efficiency based on digital evidence. However, the practical application of LIDAR, UAS photogrammetry, and continuous monitoring revealed a significant disparity between data collection and data utility. While these modalities successfully captured the "as-built" state, the high computational intensity and manual effort required to process these results for norming purposes created a feedback loop that was too slow for active site management. This technical realization—that the limits of existing digital workflows were obstructing the goal of real-time productivity measurement—serves as the primary catalyst for the broader technical framework proposed in this research.

A pivotal component of the second phase entailed the development of a pilot construction site in Sošice, Croatia, erected solely for the NORMENG project (Figures 1 and 2). This controlled environment facilitated the evaluation of methodologies for progress and quantity monitoring, alongside norm standardization, with emphasis on real-time tracking of on-site activities using various technologies like LIDAR, UASs, handheld cameras, etc. The Sošice pilot underscored both the strengths and shortcomings of these approaches in quantifying constructed volumes and formulating contemporary standards for energy-efficient construction. The initiative concluded with practical field trials in collaboration with con-

struction firms and investors, culminating in the establishment of a robust framework for resource standardization in energy-efficient building practices.



Figure 1. Pilot construction site—masonry works phase.



Figure 2. Pilot construction site—façade works phase.

2.2. AutoGreenTraC Project as Solution Developing Phase

The relationship between the NORMENG and AutoGreenTraC projects is one of sequential evolution: NORMENG established the empirical requirements for construction "norming" but exposed the critical limitations of current digital workflows, while AutoGreenTraC serves as the technological response to those limitations. Specifically, NORMENG demonstrated that while high-precision data could be captured, the latency between acquisition and reporting to stakeholders—investors and construction firms—rendered the data reactive rather than proactive. Building on these documented limitations, AutoGreenTraC was launched to develop a streamlined data flow strategy. Its primary objective is to reduce the temporal gap between site capture and actionable reporting by moving away from traditional, heavy computational processes toward the tiered edge-cloud architecture described in this framework.

While the NORMENG project confirmed the capabilities and accuracy of SfM-MVS, it also exposed its limitations regarding processing time and the inability to provide immediate feedback for active site management. AutoGreenTraC shifts the research focus from static resource standardization to dynamic, accelerated transfer of precise information from the physical site to the DT.

The project integrates advanced data acquisition technologies—including autonomous UAVs, 360-degree vision, LIDAR, and IoT sensors—with AI-driven computer vision al-

gorithms designed to automate the detection of construction elements and materials. By bypassing manual processing, the project aims to enable near-real-time reporting that supports not only schedule and cost control but also the "Green Transition" of the construction sector. Specifically, the project targets the optimization of production processes to minimize waste and the development of methodologies for tracking the carbon footprint (CO_2) of on-site activities. Through collaboration with industry partners (TVD System and PERI), the proposed methodology moves beyond theoretical modeling to validate how automated, high-velocity data pipelines can facilitate proactive decision-making and sustainable construction practices in real-world environments.

3. Problem Statement: Digitalization Expectations vs. Reality

The architecture, engineering, and construction (AEC) industry is currently in a transformational phase characterized by the need to rapidly digitize physical assets and to integrate "Construction 4.0" methodologies [4,5]. Despite these advancements, widespread adoption of digital solutions for automated progress monitoring and quality assurance remains limited [8].

The research demonstrates that construction professionals have high expectations for digitalization technologies in progress monitoring and quality reporting. Here, the focus is on ease of use, real-time monitoring, and data accuracy [13–15]. However, implementation challenges, particularly high technological costs for specialized hardware and software, complexities in managing voluminous point cloud data [16,17], and organizational resistance stemming from steep learning curves in data processing [18,19] create significant gaps between expectations and actual deployment [4,8]. The research suggests successful digitalization requires not merely technology adoption but integrated organizational change, comprehensive training, and clear value demonstration.

The analysis reveals a clear divergence in stakeholder requirements: a rigorous demand for millimeter-level precision in quality reporting, predominantly addressed by terrestrial and mobile LIDAR systems [20–22], contrasts with an escalating need for high-velocity, streamlined documentation for progress monitoring, currently dominated by 360-degree camera solutions [21,23,24]. SfM-MVS provides a hybrid approach that balances photorealistic visual fidelity with moderate geometric accuracy (typically sub-centimeter under optimal conditions), though its performance remains highly sensitive to environmental factors, such as illumination, surface texture, and image overlap, as well as processing parameters [25,26].

To resolve technical ambiguity regarding "real-time" monitoring, this framework distinguishes three distinct latency components:

- Capturing latency: The time required for physical site data acquisition. While contemporary TLS has reduced this from weeks to hours, it remains a bottleneck for daily monitoring compared to 360° cameras.
- Processing latency: The time required for data registration, denoising, and AI-driven semantic extraction. This includes the "stitching" of point clouds, which can exceed the capture time itself (e.g., 5 h of capture vs. 8 h of processing).
- Integration latency: The "Information Gap"—the time required to align processed data with BIM and update the Digital Twin for stakeholder decision-making.

Our framework specifically targets the reduction in processing and integration latency, acknowledging that capturing latency is often dictated by hardware and site accessibility.

In Section 3.1, the authors present the digitalization process that is currently the industry standard with its positive and negative sides, whereas in Section 3.2, the users' (construction industry) expectations are presented. These two aspects of the digitalization

processes are usually not well aligned when it comes to the needs of progress monitoring and immediate quality control.

3.1. Digitalization Process

3.1.1. Light Detection and Ranging (LIDAR): The Benchmark for Geometric Precision

LIDAR remains the standard for geometric accuracy in construction. Operating on the principle of emitting pulsed laser light and measuring the time-of-flight for its return, LIDAR systems generate highly dense point clouds with millimeter-level precision, capturing intricate details of "as-built" conditions [2,6].

Terrestrial laser scanning and manual photogrammetric documentation, while geometrically precise, are inherently labor-intensive—requiring multiple scan positions and setups—discontinuous, and often disruptive to ongoing site operations [27–30]. TLS has historically been a time-consuming process, often requiring several days for the comprehensive scanning of large facilities [27–29]. Solely for the Sošice experimental building, a scanning protocol took from 1 to 5 min per scanning position (Figure 3). Here, for the scanning protocol, the construction stopped for at least 15–50 min for the scanning session. On the other hand, the LIDAR scanning project shown in Figure 4 is a construction site stretching over 8000 m^2 with six partially completed houses, which took a solid 5 h of recording and more than 8 h of stitching. Nevertheless, recent technological advancements have significantly recalibrated expectations regarding the speed of data acquisition. While conventional surveying methodologies historically necessitated weeks for extensive site documentation, contemporary TLS systems are now capable of achieving equivalent data capture within a timeframe ranging from hours to a few days.

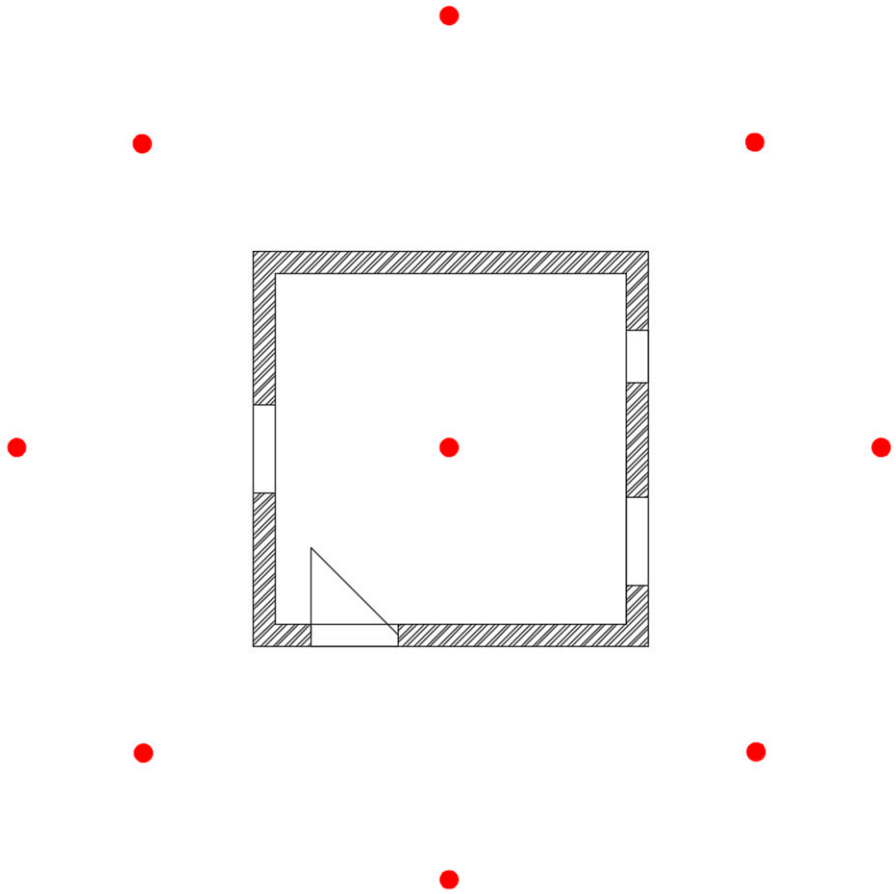


Figure 3. LIDAR scanning stations during the Sošice masonry works phase (red dots—scanning positions).

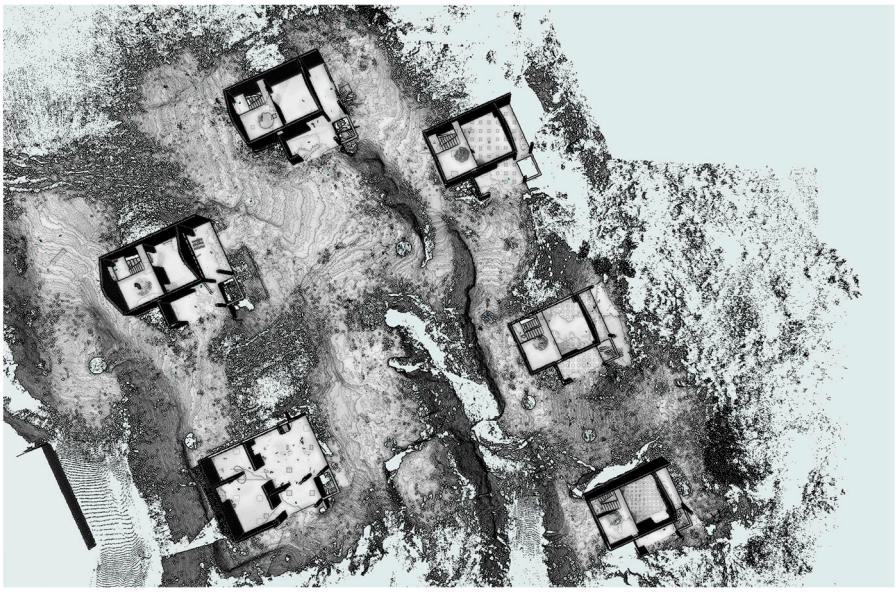


Figure 4. LIDAR scanning result on a construction site.

Despite its precision, LIDAR faces resistance due to several inherent limitations. The most significant is data heaviness [16,31]. A 20 GB file(s) for a single section is not uncommon, creating massive display and transfer issues [17]. Handling these massive point clouds requires specialized hardware and software, often creating a bottleneck where only a few "power users" can access and manipulate the data.

Another limitation is occlusion. LIDAR is a line-of-sight technology, meaning it can only measure what it can "see." Complex construction sites with clutter, equipment, and temporary structures require many scan positions (setups) to avoid "shadows" in the data, increasing field time (Figure 5). Experts often must plan scan locations meticulously to ensure complete coverage, adding to the logistical complexity of the operation [30,32].

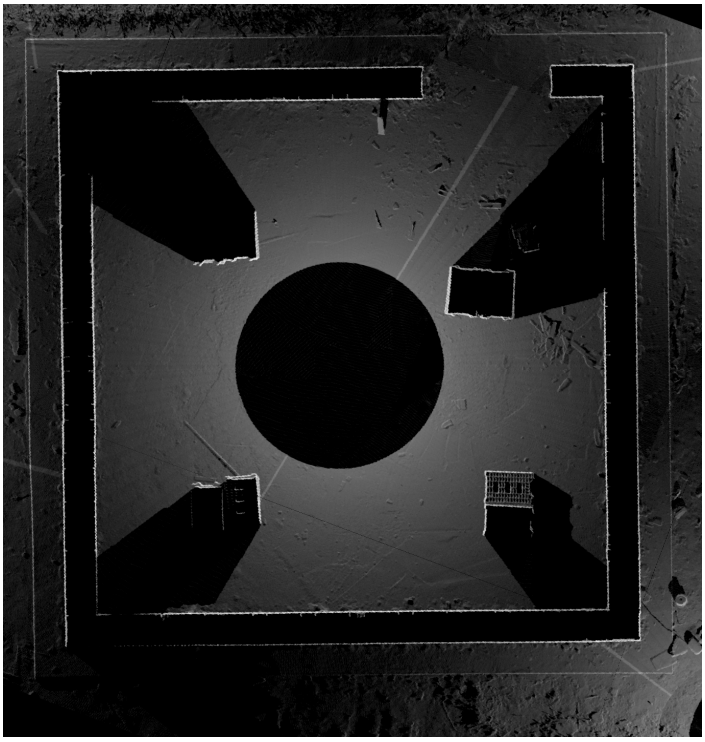


Figure 5. LIDAR scanning result—taken during the Sošice masonry works phase.

Furthermore, processing LIDAR data involves a steep learning curve. Although data acquisition has become easier, scan registration, denoising, and cleaning of point clouds, and semantic feature extraction from them remain specialized skills [18,19]. The lack of an industry-standard "undo" function in some software packages violates user expectations and complicates error correction, adding to user frustration.

3.1.2. SfM-MVS and Photogrammetry: The Visual-Geometric Hybrid

SfM-MVS algorithms enable the reconstruction of 3D geometry from 2D image sequences by first estimating camera poses and a sparse point cloud via SfM, followed by dense reconstruction via MVS [25,26,33]. This methodology has seen increased adoption, driven by the proliferation of unmanned aerial vehicles and high-quality consumer cameras, which democratize access to 3D data generation [14]. Nevertheless, for expert users, SfM-MVS occupies a specialized niche, primarily distinguished by its visual fidelity and its capacity for rapid large-area coverage, rather than its absolute geometric precision. While photogrammetry offers visual richness and broad coverage, it inherently struggles with accurately delineating model boundaries and ground surfaces, particularly impacting analyses requiring precise structural information [34,35].

The quantitative reliability of SfM-MVS exhibits considerably greater variability compared to Light Detection and Ranging, being largely contingent upon environmental factors such as illumination, surface texture, and the degree of image overlap. In controlled benchmark investigations, for instance, mapping a 20 m trench, SfM-MVS utilizing drone imagery achieved satisfactory scaling accuracy and horizontal georeferencing when supported by robust Global Navigation Satellite System data [36,37]. Nevertheless, vertical translation errors can be substantial, potentially reaching up to 8 m in the absence of established ground control points [38,39]. This intrinsic unpredictability contributes to a cautious approach among specialists, particularly regarding the exclusive reliance on SfM-MVS for critical vertical measurements or the verification of fabricated components.

The integration of LIDAR for precise geometric measurements and photogrammetry for high-fidelity visual textures is increasingly recognized as an effective hybrid approach across diverse construction monitoring applications [2,6,35]. Within this integrated framework, LIDAR instrumentation provides precise spatial measurements, forming a foundational geometric representation, while photogrammetric methods contribute rich, detailed visual information, thereby augmenting the geometric data with high-resolution texture and appearance (Figure 6).



Figure 6. SfM-MVS using 360 cameras for underground works progress monitoring.

A significant challenge for experts employing SfM-MVS pipelines is the high computational intensity of the data processing workflow. Unlike LIDAR, which directly acquires

3D point coordinates via time-of-flight laser ranging, SfM first estimates camera poses and a sparse point cloud through feature matching across images, while subsequent MVS densification derives depth maps algorithmically from pixel disparities (parallax) in numerous overlapping views [25,26,40]. This inherent computational demand is considerable [40–42]. Achieving high-fidelity densification in specialized software, such as Metashape or PIX4D, consequently mandates substantial Graphics Processing Unit resources and can incur processing durations extending from several hours to multiple days for extensive datasets.

3.1.3. 360-Degree Cameras: The Velocity of Documentation

360° cameras have emerged as a leading tool for efficient construction progress monitoring, particularly valued for their rapid panoramic image capture from single stations despite introducing higher noise and variability in 3D models compared to terrestrial laser scanning [43,44].

In this chapter, 360° camera devices are considered as data-capturing devices for 360° SfM-MVS reconstruction, where 360° videos or photos are used for the structuring of sparse or dense point clouds using SfM-MVS protocols. Besides these, 360° cameras can also be used for photographic documentation without any complex processing; they are mainly used for immediate processing of images or within SLAM-based mapping. These, however, are not in the scope of this review.

Although these cameras do not offer the metrological precision characteristic of LIDAR systems, their rapid data acquisition capabilities, operational simplicity, and cost-effectiveness render them exceptionally suitable for the frequent documentation requirements prevalent in general contracting. This technology has fundamentally redefined expectations regarding "site visibility," transitioning industry practices from infrequent photographic documentation to extensive, immersive virtual site inspections.

The primary driver for adopting 360° cameras is their substantially reduced data acquisition time compared to conventional multi-view photogrammetry, which requires numerous overlapping images from multiple viewpoints, and terrestrial laser scanning, which necessitates multiple stationary positions to ensure comprehensive coverage and mitigate occlusions [30,32,43,44]. Empirical investigations demonstrate that 360-camera solutions can expedite site documentation by a factor of five relative to other common methods and achieve a tenfold increase in speed over manual photographic processes (Figure 6).

This efficiency stems from the device's capacity to acquire full 360° panoramic imagery from a single vantage point, thereby substantially minimizing the requisite number of acquisitions for comprehensive site coverage [44]. Despite their operational simplicity and elevated data density, these cameras frequently yield 3D models exhibiting elevated noise levels and geometric variability relative to terrestrial laser scanning and multi-view photogrammetry [43]. Nevertheless, recent innovations—including AI-augmented 360° cameras leveraging wall texture classification, image segmentation, and feature-extraction algorithms—have yielded enhanced accuracy and reliability for automated construction progress monitoring [23].

3.2. User Expectations

In the context of progress monitoring, quality control, and productivity assessment, the expectations from the digitalization of construction processes often diverge significantly from the observed realities, particularly due to challenges in data acquisition, processing, and integration for actionable insights. Contemporary expert practitioners increasingly prioritize integrated ecosystems that facilitate a seamless continuum of data capture, analytical processing, and actionable output, moving beyond the demand for isolated, standalone

tools. Investigations into user expectations consistently underscore pivotal requirements that extend beyond mere hardware specifications, with a pronounced emphasis on interoperability, user experience, and process automation [45–47].

3.2.1. The "Single Source of Truth" Expectation

Users are fatigued by data silos where information is trapped in proprietary formats or disconnected platforms. The expectation is that reality capture data (point clouds/meshes) must integrate directly into the Common Data Environment (CDE) to enable interoperability [48–50] and clash detection [49,50].

In this framework, the single source of truth is not merely a collaborative goal but a systemic architecture built upon the ISO 19650-2 [51] framework for a Common Data Environment (CDE). It utilizes a data-centric model where reality capture data and BIM objects are linked via unique identifiers (GUIDs), ensuring data is stored exactly once to prevent redundancies. Governance is enforced through automated status gateways that manage the transition of data from "captured" to "verified," providing an immutable provenance that manual, document-centric methods lack. This architecture ensures that "firm" information is always presentable to stakeholders, as every reported metric is directly traceable to a verified dataset within the CDE.

3.2.2. Quick Access to Information and Proactive Quality Reporting

Construction projects are burdened by complex activity interactions that often lead to delays and increased costs. Consequently, there is a growing expectation for artificial intelligence to automate the labor-intensive data processing workflows, thereby enhancing productivity [47] and enabling experts to focus on advanced analytics and strategic decision-making [46,47]. This automation would mitigate the time-consuming and labor-intensive nature of manual data analysis, which often involves organizing and localizing raw visual data [8], but the requested information often comes with a lag, impeding real-time adjustments and proactive interventions necessary for efficient project management [49].

3.2.3. Ease of Use and Precision of the Information

The primary challenge lies in the current inability to seamlessly and automatically translate complex, dynamic site conditions into structured, actionable data for immediate assessment of progress and quality, leading to reactive instead of proactive project management [52,53].

Many studies have focused on applying novel technologies like laser scanning, computer vision, and radio frequency identification to achieve real-time automated monitoring and control, yet significant challenges persist in their practical application within construction projects.

4. Technological Advancements to Data Capture and Analysis

Rather than suggesting the replacement of existing hardware, this section evaluates the specific processing requirements of currently deployed sensors to determine how they must be reconfigured within a near-real-time control loop. When it comes to tracking the construction process in the digital surroundings (recorded construction), progress monitoring depends exclusively on documented evidence of completed work, whereas quality control during the construction process relies on the assessors' ingenuity in exploiting available technologies.

The present study focuses on advancements in monitoring:

- Structural element completeness—automated verification that physical components (e.g., columns, beams, or masonry walls) have been completed as per design.

- Material placement verification—confirming that specific materials are correctly positioned according to design specifications.
- Construction accuracy and alignment—ensuring that the physical structure matches the geometric design.
- Surface quality assessment—the inspection of the "as-built" finish, such as checking for cracks, flatness, or curing quality.
- Progress quantification metrics—assigning measurable units, such as cubic meters of concrete or square meters of masonry finished, to the raw recorded construction site data.
- Real-time defect detection—detection of construction processes that may cause defects (i.e., wrong placement of formwork) or detection of faults in the construction process (i.e., missing connection elements in the wall formwork or missing supporting elements in ceiling formwork).
- Productivity metrics—tracking the efficiency of labor and machinery in real time, providing data-driven insights into the velocity of construction activities.

This section presents an exhaustive review of the state-of-the-art technological advancements engineered to dismantle these bottlenecks. The analysis is bifurcated into two primary domains: (1) technologies that accelerate the physical collection of field data, moving towards autonomous, ubiquitous, and continuous capture workflows; and (2) technologies that accelerate the cognitive processing of this data, leveraging advanced artificial intelligence (AI), neural rendering, and edge computing to transform raw sensor inputs into actionable semantic understanding in near real time. By examining these domains, we establish the technological mandate for a modernized, data-driven construction control loop.

4.1. Technologies to Speed Up Data Collection

4.1.1. Autonomous Mobile Robotics (AMR) and Quadruped Systems

Mobile ground robots, particularly quadruped platforms such as Boston Dynamics Spot and ANYbotics ANYmal, have emerged as promising solutions for autonomous, repeatable reality capture on active construction sites. These platforms offer several advantages over traditional manual inspection: they can navigate complex terrain, operate in hazardous environments, carry multiple sensor payloads, and perform scheduled autonomous missions without direct human supervision.

Yang et al. [52] proposed a cyber-physical systems framework for construction progress monitoring, employing mobile robots as primary data collectors. It integrates robotic platforms into ecosystems, combining reality capture, cloud processing, and automated analytics. This work established principles for robot-enabled continuous monitoring.

Halder et al. [54] developed a quadruped robot system for real-time remote construction progress monitoring with AR features. It streams live 360° images to operators for AR-based as-built vs. DT comparisons. This advances near-real-time inspections, but high-resolution data strains processing bandwidth. Building on this foundation, Halder et al. [54] developed a robotic cyber-physical system for automated reality capture and visualization. It integrates quadruped robots, cloud processing, and AR interfaces. Their 2024 study advances autonomous mission planning and data pipelines, enabling regular scans with minimal human input. Challenges persist in dynamic obstacle avoidance and coverage optimization.

Zhao and Cheah [55] created a BIM-enabled robotic system for indoor construction progress monitoring. It uses BIM models to generate waypoint navigation paths for repeatable, schedule-aligned scans. Integrating SLAM with BIM navigation advances practical robotic monitoring. Chen [56] examined rapid robotic scan-to-BIM workflows using

LIDAR-equipped mobile platforms and AI. Robotic scanning with automated semantic parsing reduces manual processing time by 60–70% compared to traditional terrestrial laser scanning, though accuracy trade-offs require management.

The core advantage of robotic inspection lies in the transition from manual, discretionary data collection to automated, repeatable missions. The recent literature quantifies these benefits: quadruped robots reduce repetitive site inspection workloads by 25–90%, depending on mission complexity and autonomy level [57].

4.1.2. Unmanned Aerial Systems (UASs) and Beyond Visual Line of Sight (BVLOS)

Unmanned Aerial Systems (UASs) have evolved from experimental platforms to indispensable tools for construction site surveying, earthworks volumetrics, and structural inspections. State-of-the-art developments have advanced beyond manual piloting to fully autonomous "drone-in-a-box" systems and beyond visual line-of-sight operations, substantially enhancing the frequency, coverage, and scalability of aerial data acquisition [58,59].

Huang and Jiang [59] reviewed UAS applications in construction management from 2016 to 2021, analyzing over 150 papers to identify trends, uses, and gaps. They found progress monitoring and inspection to be the most advanced UAS applications, with benefits in coverage speed, safety, and data quality.

Ibrahim et al. [60] optimized UAS reality capture plans to improve photogrammetric quality while reducing flight time and constraints. Their multi-objective framework accounts for coverage redundancy, viewing angles, ground sampling distance, and line-of-sight needs to produce Pareto-optimal plans. Vrachnos et al. [61] developed a framework integrating UAS capture, neural rendering, and DT for vision-based 3D inspections. It supports high-fidelity remote checks of building façades and infrastructure, with automated 3D defect localization at centimeter-level accuracy for access-free maintenance planning. Samsami [58] systematically reviewed automated construction inspection and progress monitoring across 200+ papers, covering capabilities, domains, and barriers. UASs perform best on large outdoor projects but face challenges indoors due to GPS limitations and collision risks.

UAS photogrammetry enables rapid horizontal site coverage. A single flight can map hundreds of acres, yielding high-resolution orthomosaics and digital surface models for comprehensive progress views. This excels for earthwork volume tracking, supplanting slow, hazardous manual surveys. Here and in other tasks, "drone-in-a-box" systems boost autonomy via weatherproof stations with automated battery charging/swapping. Nevertheless, regulatory limitations may be restricted here.

Advancements like onboard Edge AI help cut cloud-upload latency for gigabyte-scale imagery. Therefore, flight-time inference detects objects/defects, transmitting compact insights (e.g., site logistics maps and safety alerts) instantly for real-time decisions.

4.1.3. Continuous Monitoring via IoT and Fixed Visual Sensors

The integration of Internet of Things devices and fixed visual sensors offers a persistent data stream, enabling real-time insights into construction progress without the constraints of mobile platforms. Internet of Things (IoT) sensor networks provide continuous, automated data streams that complement periodic visual and geometric capture. While cameras and scanners provide rich spatial information, IoT sensors monitor temporal processes and conditions: concrete curing, structural vibration, environmental parameters (temperature, humidity), equipment operation, worker safety, and resource consumption. The integration of IoT with BIM and AI enables real-time quality reporting, predictive analytics, and automated alerting that support proactive project management.

Han et al. [62] presented a BIM-IoT integrated framework for advanced road compaction quality monitoring and management. Their system combines GPS-equipped compaction rollers, embedded sensors, and BIM models to provide real-time visualization of compaction coverage and quality, with automated detection of under-compacted areas and corrective recommendations, reducing rework by 35% in field trials. Wang et al. [63] developed a real-time monitoring system for fresh concrete vibration quality using CNNs and IoT. It employs wireless sensors on concrete forms with edge-deployed CNNs for real-time classification (94% accuracy), enabling immediate corrections and reducing defects by 28% versus manual inspection.

Abdul Nifa and Teong [64] proposed a Digital Twin-based method integrating IoT sensors, BIM, and 4D schedules for dynamic project performance monitoring, supporting predictive analytics and improved schedule accuracy. Pučko et al. [65] demonstrated that continuous low-precision 3D scans from active workplaces outperform periodic high-precision scans for progress monitoring. AI-enhanced IoT sensors further enable real-time monitoring of structural health, energy consumption, and environmental conditions, facilitating predictive maintenance and resource optimization [66].

The most robust monitoring employs sensor fusion of visual and non-visual IoT data for cross-validated streams. For example, concrete maturity sensors confirming slab strength can align with visual verification of formwork removal, enhancing DT reliability as a real-time single source of truth [67,68].

4.2. Real-Time Construction Quality Reporting

Acquiring comprehensive, high-fidelity data constitutes only half the challenge in digital construction monitoring. The second primary bottleneck entails the expeditious processing, analysis, and interpretation of this multimodal data to yield actionable insights that inform decision-making. Unprocessed point clouds, imagery, and sensor streams must be converted into quantifiable progress metrics, quality assessments, geometric deviation reports, and predictive alerts. This section reviews technologies that expedite data analysis and afford near-real-time comprehension of on-site conditions.

4.2.1. From Photogrammetry to Neural Radiance Fields (NeRF) and 3D Gaussian Splatting (3DGS)

3D reconstruction from 2D construction site imagery has relied on the SfM-MVS pipeline, which generates sparse point clouds via feature matching followed by dense point clouds or meshes via multi-view disparity estimation. SfM-MVS remains computationally intensive, often requiring hours or days to process large-scale datasets. Moreover, it struggles with featureless regions or reflective/transparent surfaces prevalent in construction environments, yielding reconstruction artifacts and holes [69,70]. A significant paradigm shift is underway with neural rendering techniques.

Neural Radiance Fields introduced a paradigm shift by representing scenes via multilayer perceptrons that parameterize a continuous 5D radiance field—spatial position and viewing direction—rather than discrete meshes or point clouds. NeRFs excel at synthesizing photorealistic novel views via volume rendering and capturing complex view-dependent effects (e.g., specular reflections, transparency) that elude traditional photogrammetry. However, the original NeRF was prohibitively slow to train and render. Recent optimizations, such as Instant Neural Graphics Primitives and NeRFacto, have reduced training times from hours to seconds or minutes, making them viable for rapid site visualization [69–71].

3D Gaussian Splatting (3DGS): The most recent advancement in reality capture is 3D Gaussian Splatting (3DGS). Unlike NeRFs, which require querying a neural network for every pixel during rendering, 3DGS represents the scene as a collection of millions

of 3D Gaussians (anisotropic ellipsoids) that carry color and opacity information and are rasterized directly to the screen [72,73].

4.2.2. AI and Machine Learning for Construction Monitoring

AI and ML are the transformative advancements in automating the analysis of multi-modal construction site data. Deep learning models, such as convolutional neural networks, facilitate the automated detection, classification, and quantification of construction elements and activities using image analysis and point cloud segmentation [74]. These approaches process data orders of magnitude faster than manual methods while attaining equivalent or superior accuracy, thereby enabling near-real-time progress monitoring and quality assessment [58].

Modern systems employ state-of-the-art deep learning architectures for visual and 3D data, including object detection networks for identifying construction elements, equipment, and workers in images and video [74], semantic segmentation models for pixel-wise or point-wise classification of images and point clouds in rapid scan-to-BIM workflows [74], instance segmentation for delineating individual object instances to support accurate counting and measurement [58], and transformer-based models that demonstrate improved performance on complex scenes with occlusion and clutter, advancing beyond traditional CNNs [58].

Key approaches include end-to-end automated pipelines that integrate data capture, AI processing, and reporting into unified cyber-physical systems, minimizing manual intervention. Pioneering examples include Golparvar-Fard et al.'s [60] framework, which fuses visual/sensor data via deep learning for element detection/classification and automated schedule-aligned reporting; Pour Rahimian et al.'s [75] gamified BIM-ML system for on-demand monitoring via imagery/point cloud analysis matched to BIM objects; and Abreu et al.'s [74] BIM-AI Digital Twins with real-time object detection and point cloud segmentation for continuous updates. These pipelines automatically ingest new data, run inference, update the DT, and generate stakeholder reports, as validated across 200+ studies [58].

Transfer learning and domain adaptation address data scarcity and poor generalization—primary barriers beyond detection tasks [58]—by pre-training on massive general-domain datasets, then fine-tuning with limited construction data, cutting training requirements and elevating performance across sites.

Multimodal learning combines imagery, point clouds, BIM models, and IoT sensors in unified deep learning models, improving robustness, contextual understanding, and accuracy in complex environments [58,67,68]. It leverages complementary strengths, such as visual textures from imagery and precise metrics from sensors (e.g., concrete maturity), as in sensor fusion for cross-validated DT [67,68].

Human-in-the-loop and active learning overcome data scarcity, generalization issues, and interpretability by engaging experts for AI output validation. Feedback improves models, with active learning prioritizing uncertain cases [58]. For instance, Pour Rahimian et al.'s [75] system elevated accuracy from 78% to 94% via stakeholder input, with analogous refinements in Golparvar-Fard et al. [60] and Abreu et al. [74].

4.2.3. Edge AI and On-Device Processing

Three aspects are identified as the key ones: latency elimination and bandwidth efficiency, on-device photogrammetry and SLAM, and hybrid edge-cloud architectures.

By executing optimized inference models (Edge AI) directly on local edge hardware, the framework reduces latency for safety-critical tasks like worker helmet detection to approximately 30–50 milliseconds per frame. This allows the information on the critical

issue to be transferred almost instantaneously, bypassing round-trip data transmission delays [58]. This empowers instantaneous safety interventions critical to construction sites, such as real-time detection of worker falls, PPE non-compliance, or crane proximity hazards using edge-enabled cameras, which can immediately activate alarms and avert accidents [58,74]. Moreover, by preprocessing and filtering data at the source—uploading only compressed metadata, alerts, or key events rather than full raw video feeds—edge AI achieves dramatic bandwidth reductions (often >90%) essential for unreliable, low-connectivity environments prevalent in remote or urban construction settings [58].

Hardware advancements in mobile SoCs and AI accelerators enable the execution of lightweight photogrammetry and visual SLAM algorithms directly on edge devices such as smartphones, drones, and site cameras [58]. Although on-device outputs typically yield lower resolution and density compared to cloud-based counterparts, they deliver critical real-time capabilities: immediate verification of scan coverage, preliminary 3D reconstructions for basic progress monitoring, and on-site quality checks before operators depart—averting costly revisits and enabling proactive adjustments in dynamic construction environments [58–74].

Hybrid edge-cloud architectures distribute computing tasks across hierarchical tiers—on-device, edge servers, and cloud data centers—to optimize latency, bandwidth, privacy, and scalability [58]. Time-critical inferences, such as real-time safety monitoring (e.g., worker falls, PPE non-compliance, or equipment hazards) and preliminary progress monitoring, execute locally to enable instantaneous responses and eliminate transmission delays [58,74]. Conversely, resource-intensive tasks like detailed 3D reconstructions, multimodal data fusion, or long-term analytics are offloaded to the cloud for superior computational power and storage [58,67,68].

Emerging developments in hybrid frameworks, federated learning for decentralized model training without raw data sharing, and microservices for modular, scalable deployments have propelled responsive construction monitoring systems, as evidenced in integrated pipelines fusing edge AI with cloud BIM updates across diverse sites [58,74,75].

4.2.4. BIM Integration and Digital Twins

Building Information Modeling furnishes structured semantic, geometric, and spatiotemporal context—encompassing as-planned models with element hierarchies, precise geometries, and schedule linkages—that converts heterogeneous raw construction data streams into coherent, actionable project intelligence [58]. Digital Twins extend BIM by instantiating dynamic, perpetually synchronized virtual replicas of physical construction states, assimilating multimodal real-time data from imagery, point clouds, sensors, and IoT [58,74,75]. Contemporary research outlines key advancements, including automated Scan-to-BIM workflows via point cloud segmentation and procedural modeling [58,74,76], BIM-guided data acquisition with optimized robotic paths [58,60,74], and extensible DT platforms enabling predictive analytics, discrepancy detection, and scenario simulation [58,75].

A primary innovation lies in automated Scan-to-BIM workflows [74,76], which integrate point cloud processing from laser scans [76], AI-based semantic segmentation [75], and procedural geometric modeling [74] to generate accurate as-built BIM models [58]. Studies confirm these reduce manual effort by 50–70% versus traditional methods [58], enabling seamless updates to the DT as in Pour Rahimian et al.'s BIM-ML system [75] and Abreu et al.'s segmentation pipelines [74].

Building on this, BIM-guided data collection leverages BIM models to automatically generate optimized capture missions for robots, drones, and scanners, minimizing coverage gaps, redundant scans, and occlusions through path optimization and viewpoint

planning [58,60,74]. By providing semantic contexts—such as element locations, visibility constraints, and required inspection points—BIM improves capture planning, automates data interpretation and registration to as-planned models, and supports seamless DT updates [58,75,76].

Furthermore, semantic data linking to BIM enables automated mapping of detected elements and sensor measurements to corresponding BIM objects. This helps spatial awareness, context-aware analytics, real-time detection of discrepancies between as-planned and as-built conditions, and proactive quality assurance [58,75,76].

In addition, 4D/5D DT integration combines geometric BIM with time and cost dimensions to create comprehensive, dynamic DT that supports predictive analytics, scenario simulation, and resource optimization [58]. By linking as-planned DT models with real-time sensor data—such as scans, IoT measurements, and AI-detected deviations—these twins enable continuous re-forecasting of schedules and budgets, facilitating proactive adjustments to mitigate delays, cost overruns, and discrepancies [58,74,75,77].

Finally, cloud-based collaborative platforms provide scalable, web-based DT interfaces that enable multi-stakeholder access to fused project data from the DT, sensors, and AI analyses, facilitating real-time distributed decision-making, collaborative oversight, and on-demand monitoring [58,74,75]. These platforms leverage cloud infrastructure for seamless scalability, secure data sharing without raw data exposure, robust storage, and advanced cross-project analytics—enabling performance benchmarking, predictive insights, and knowledge transfer across sites [58,74,75].

5. Discussion and Challenges: AutoGreenTraC Research Framework

While the literature demonstrates that individual data acquisition technologies—specifically UAS photogrammetry, LIDAR, and 360° vision—have reached technological maturity, their isolated application fails to address the construction industry's urgent need for real-time optimization. The review exposed the critical gap: latency between data acquisition and the generation of actionable insight. Bridging this gap is not merely a technical convenience but a prerequisite for the AutoGreenTraC project goals.

The literature review reveals three fundamental bottlenecks that align directly with the AutoGreenTraC project's core mission of accelerating information transfer from physical construction sites to digital environments:

- Data acquisition speed vs. accuracy trade-offs.
- Computational latency lacking near-real-time feedback.
- Integration gaps and a single source of truth deficit.

Building on the literature synthesis, a conceptual framework for AutoGreenTraC was structured around three interconnected pillars that directly respond to identified technological gaps:

- Autonomous data acquisition.
- Accelerated processing through edge computing.
- Semantic integration, DT, and actionable intelligence.

The literature review demonstrates that single-modality techniques are insufficient for thorough construction monitoring. Using autonomous mobile robots cuts repetitive inspection tasks by 25% to 90% and boosts safety by slashing human exposure to hazards by 72% [58]. BVLOS drone flights can reduce data collection time by up to 90% compared to ground-based approaches [60]. IoT sensor networks complement these by enabling continuous temporal monitoring of processes imperceptible to visual sensors, such as concrete curing, structural vibrations, and equipment operations [67,68]. AutoGreenTraC must design an integrated data acquisition strategy that:

- Deploys autonomous UASs and/or ground robots for scheduled, repeatable site documentation.
- Integrates 360-degree cameras and surveillance systems for continuous visual monitoring.
- Embeds IoT sensors for process-level insights.

Attaining near-real-time responsiveness in construction monitoring necessitates a profound reconfiguration of processing architectures, as underscored by systematic reviews identifying computational latency as a critical bottleneck [58]. Scholarly emphasis on edge AI and revolutionary neural rendering innovations—chief among them 3D Gaussian Splatting (3DGS)—provides a robust pathway for the AutoGreenTraC computational paradigm, building on AI-driven semantic segmentation and BIM integration advances [74,75]. Unlike established SfM-MVS or NeRF workflows, which impose substantial processing delays ranging from minutes to hours [58], 3DGS enables real-time rendering of photorealistic 3D scenes by explicitly modeling them as millions of rasterizable Gaussian primitives, achieving speeds orders of magnitude faster while maintaining high fidelity. Consequently, AutoGreenTraC's theoretical formulation must incorporate a tiered processing hierarchy:

- First tier: Lightweight CNNs or IoT analyses performed locally and continuously to perform immediate reporting and warning. Reporting output includes safety alerts, preliminary progress verification, and basic element counting.
- Second tier: More sophisticated models run on construction site edge servers for tasks requiring moderate computational resources. Reporting outputs include material classification, element detection and segmentation, preliminary scan-to-BIM alignment, and 3DGS-based rapid visualization. This tier targets near-real-time feedback within hours from the work being recorded with sufficient accuracy for daily decision-making.
- Third tier: High-fidelity processing occurs asynchronously for tasks demanding maximum accuracy. Reporting outputs include detailed point cloud registration and meshing, comprehensive DT updates, long-term analytics, multimodal data fusion, and 4D/5D DT maintenance. This tier operates on longer timescales (days of processing) but provides the single source of truth for project-wide coordination and the as-built documentation.

This proposed architecture directly responds to the literature's advocacy for hybrid edge-cloud frameworks while acknowledging that AutoGreenTraC's green transition goals—particularly CO_2 footprint tracking—demand both real-time feedback and comprehensive analysis.

Raw sensor data has limited value without semantic context and automated translation into project-relevant insights. The literature consistently highlights automated scan-to-BIM [58,74,76], DT-guided capture planning [60,75], and 4D/5D DT integration [58,67,75] as essential for transforming data into actionable intelligence. These approaches enable precise progress monitoring, reduce discrepancies between as-planned and as-built models by up to 90% [58], and support predictive analytics for optimized decision-making [75]. Thus, the AutoGreenTraC framework must be implemented, which includes:

- Automated Element Detection and Semantic Linking.
- Material and Quantity Tracking for Carbon Accounting.
- DT-Guided Mission Planning.
- 4D/5D DT with Predictive Analytics.

This pillar directly addresses the critical single source of truth deficit identified in the literature [58,76], alongside the industry's pressing demand for seamless integration from data capture to actionable output, which enables up to 90% reductions in as-planned vs. as-built discrepancies [58] and empowers advanced decision-making tools [75].

6. Conclusions

The construction sector continues to grapple with substantial cost overruns and schedule slippages, primarily attributable to the delays and inefficiencies inherent in conventional manual methods and nascent digital approaches to progress monitoring and quality assurance. This study bridges this pivotal shortfall through an exhaustive review of cutting-edge vision-based sensing modalities—namely, Light Detection and Ranging, SfM-MVS, and panoramic (360-degree) imaging—to formulate a robust theoretical foundation for near-real-time automated project monitoring.

The review illuminated a fundamental tension between the metrological exactitude demanded for stringent quality reporting and the elevated data throughput essential for real-time progress monitoring. LIDAR still holds supremacy in geometric precision, but the overwhelming data volumes, vulnerability to occlusions, and excessively prolonged processing times render it impractical for frequent site surveillance. On the other hand, 360-degree cameras furnish the crucial speed and cost-effectiveness essential for routine captures, albeit sacrificing the precision demanded for high-stakes applications. SfM-MVS delivers a compelling balance of photorealistic quality and geometric reliability yet persists as hindered by exorbitant computational demands and persistent inaccuracies in vertical measurements.

The principal advancement herein lies in distilling these insights into an advanced theoretical construction that prescribes the synergistic, AI-orchestrated deployment of these heterogeneous data sources. Such integration is indispensable for surmounting the intrinsic constraints of monolithic technologies. Harnessing artificial intelligence and computer vision to streamline multi-modal data processing and analysis, the proffered paradigm converts unprocessed site data into timely, decision-enabling intelligence. This evolution heralds a transformative shift from retrospective record-keeping to proactive project management, mitigating the profound economic toll of rework and temporal overruns.

Moreover, the study outlines a technical roadmap for advancing the construction industry's green transformation. By furnishing the requisite data cadence and accuracy, the framework facilitates continuous monitoring and refinement of operational workflows, which are pivotal for curbing waste and precisely quantifying on-site carbon emissions.

Subsequent investigations will prioritize empirical corroboration and deployment of this amalgamated AI-centric workflow within authentic construction settings. The main objective remains to substantiate, through rigorous testing, the framework's capacity to furnish the anticipated near-real-time feedback, thereby fostering a more streamlined, sustainable, and analytically informed paradigm for construction.

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